Applied Optoelectronics (AAOI)

Full investment research | Model assumptions and source evidence

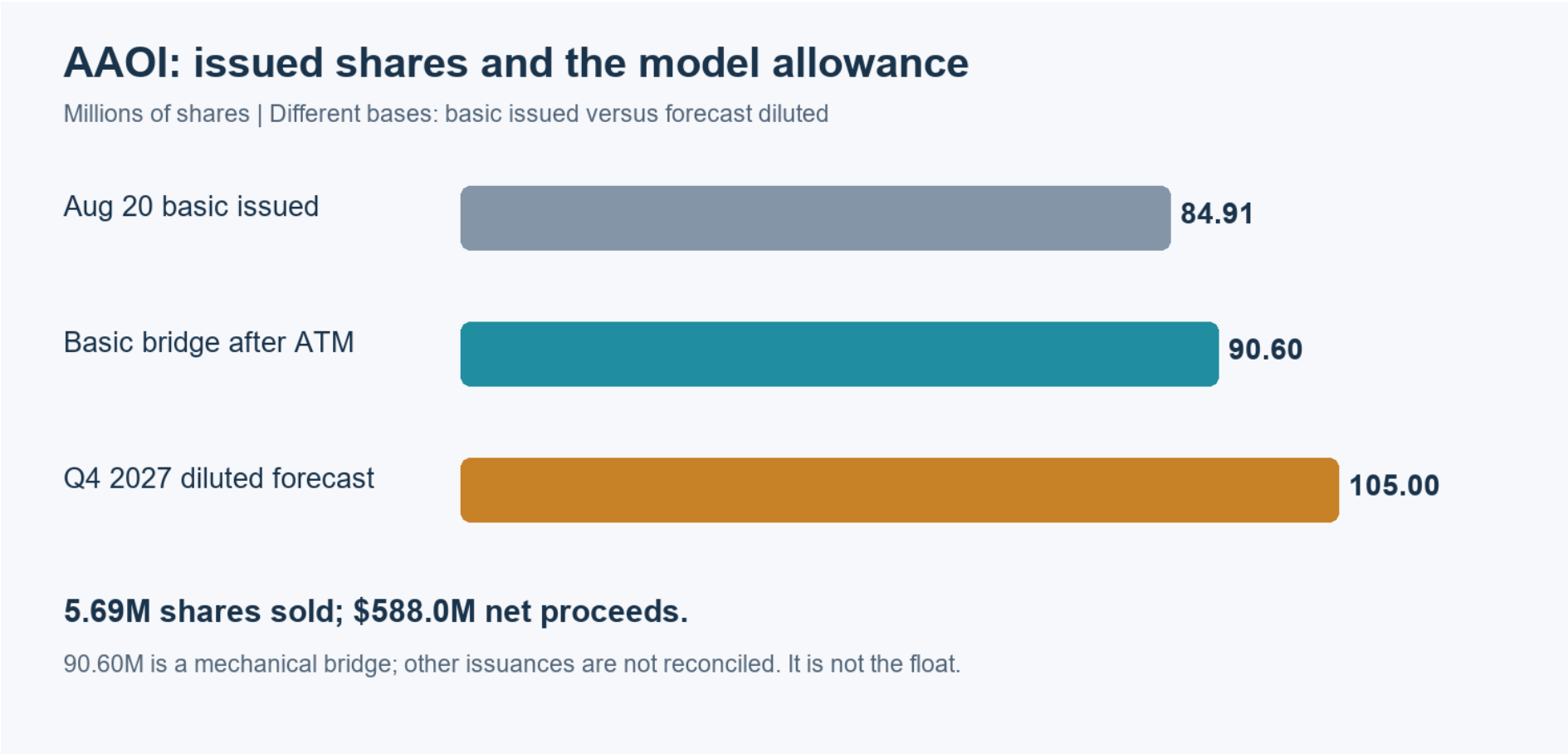
CURRENT EDITION - 6 October 2026 | Framework 8.0 | Capital instruments reconciled; operating assumptions retained

Full investment research, current probability-weighted operating model and analytical rescore. Financial assumptions, source evidence and forward estimates are distinguished throughout.

Capital and dilution: reconciled through October 6, 2026

Q4 2027 price target remains USD 463.058351; analytical Base 72 and protected Excel Base 72 remain unchanged. The saved model uses 20x forward-four-quarter operating profit and 105.0M economic/diluted shares at Q4 2027. This reconciliation supports retaining that total allowance. It does not turn a future denominator into a current legal share count or a calculated GAAP diluted weighted average.

The October 5 issuer release reports completion of the $600M ATM announced August 21: **5,694,845 shares**, a rounded average sale price of **$105.36**, and **$587,999,985.89 net proceeds**. Completion is a reported fact, recognized at 100%. The rounded average is not an exact gross-proceeds reconciliation. The latest exact company-wide common count located is **84,906,289 as of August 20**, from the August 21 prospectus. Adding the completed offering once gives **90,601,134**, a mechanical bridge that excludes any other net issuance after August 20. The October 5 Form 144 repeats the earlier August 3 count of 84,569,237; that stale field is not a new issuer count. Neither count is a float estimate. [AS1-AS3, AS12]



Capital bridge: completed ATM shares remain within the existing total-share forecast. The instrument table below supplies the October 6 reconciliation; this original exhibit is not a current legal-share or float estimate.

Capital item	Shares M	Treatment
Last exact issuer-disclosed common outstanding	84.90628900	issued common; not float
Completed August 21-October 5 ATM	5.69484500	issued common; add once
Mechanical common bridge	90.60113400	not a certified October 5 company total; other net issuance not separately disclosed
RSU shares issuable on vesting	1.09005500	excluded from August 20 common; do not also add June reserve

Capital item	Shares M	Treatment
PSU shares disclosed as issuable on performance conditions	0.62946300	disclosed count; award payout basis not expressly labeled target versus maximum in prospectus
Remaining 2030 notes, ordinary physical conversion	2.88431837	$124.925M principal / $1,000 × 23.0884; not carrying value or market value
Amazon maximum warrant reservation	7.94539900	full 10-year reservation, including initial 1.324233M; unvested portion not immediate legal common
Gross identified potential before future grants or financing	103.15036937	reserve envelope, not GAAP diluted weighted-average shares
Additional PSU sensitivity if disclosed number were target and payout reaches 200%	0.62946300	conditional conservative sensitivity, not certified extra outstanding awards
Gross envelope with the additional PSU sensitivity	103.77983237	no future discretionary equity or make-whole increment
Proposed future discretionary award allowance, Aug20 2026-Q4 2027	1.00000000	analyst reserve; range 0.5-1.5M; no claim of granted or vested shares
Envelope including PSU sensitivity and central future-award allowance	104.77983237	adequacy diagnostic, not an independently derived active denominator
Retained Q4 2027 model denominator	105.00000000	existing economic/diluted-share convention retained
Residual above the conservative envelope plus central new-award allowance	0.22016763	not a financing commitment or liquidity reserve

Issued shares and reservations are separate. The August 20 common count already incorporates the prospectus's 332,428 post-June shares sold under the earlier May ATM and 188,000 shares issued on RSU vesting. It also already incorporates prior convertible settlements, including 1,487,874 shares issued in the December 2024 exchange, 239,404 shares when the last old 2026 notes converted in July 2025, and 1,731 shares for the $75,000 conversion in Q2 2026. None is added again. The prospectus expressly excludes its 1,090,055 RSUs and 629,463 PSUs from the common anchor. These components have different latest disclosure dates; the envelope is not a single-date capitalization certificate. Later vesting or conversion transfers the relevant reserve into common and must not increase both buckets. An updated filing should roll down settled awards, warrants or principal as it rolls up issued shares. June award balances, historical grants, plan availability and outstanding August awards are not stacked. The proxy's December 2025 equity-compensation table consists of RSU/PSU rights with zero exercise price; no separate outstanding employee-option increment was identified. [AS1, AS2, AS7]

The 2030 convertible bridge is principal-based. Remaining principal of $124.925M divided by $1,000 and multiplied by 23.0884 produces **2.88431837M ordinary physical-conversion shares**, before fractional settlement and any later undisclosed changes. Its $129.142M accounting carrying value and $445.1M disclosed fair value are not conversion principal. The old 2026 notes are fully retired. The issuer can settle the 2030 notes in cash, shares or a combination. Before October 15, 2029, early holder conversion has contractual conditions; a single high stock quote is insufficient. Issuer redemption begins January 15, 2027 subject to the specified price test. A qualifying call or fundamental change can add make-whole shares: the contractual maximum rate of 29.4375 implies **3.67747969M total**, or **0.79316132M above ordinary conversion**. That is a special-event maximum, not the ordinary share forecast. For comparison, the January 15, 2027 table adds only 0.07750347M shares at a $100 stock-price input or 0.00953178M at $200; no additional shares apply above the table's $350 upper stock price. The valuation target is not an

observed stock-price input. [AS1, AS5]

Amazon vesting follows customer purchases, not total AAOI sales. The ten-year warrant reserves up to 7,945,399 shares: 1,324,233 vested initially and 6,621,166 are conditional on purchases, with full vesting after $4B of qualifying gross payments by or on behalf of Amazon and affiliates, subject to the agreement. Intermediate payment thresholds and share increments are redacted. The full reservation is included once in the conservative envelope above; it is not a claim that all shares are presently vested or outstanding. The latest 10-Q gives a $23.6956 exercise price while the original executed exhibit says $23.6954. The $0.0002 discrepancy remains identified rather than silently resolved. [AS1, AS4]

The H1 2026 filing reports $84.8M of warrant-associated revenue versus $189.066M of company datacenter revenue, a 44.85% reference ratio. A timing check therefore allocates **25% / 45% / 65% of the existing modeled transceiver pool**, with **25% / 50% / 25% subjective sensitivity weights**. The pool excludes CATV, Other and ELSFP; no customer sales are invented by assigning Amazon all company revenue. It is an allocation proxy, not a verified Amazon share of every product. Starting receipts are proxied by $33.1M FY2025 plus $84.8M H1 2026 associated revenue, and subsequent receipts lag sales by one quarter. Because revenue is not cash paid and the contractual vesting steps are redacted, the linear formula below is a bounded research diagnostic, not a legal vesting schedule or a second active dilution model:

Gross warrant proxy = 1.324233M + 6.621166M × min(cumulative qualifying-payment proxy / $4,000M, 1).

Amazon share of modeled transceiver pool	Subjective weight	Q4 2027 cumulative payments proxy, USD M	Q4 2027 gross warrant proxy, M
25%	25%	1119.953	3.178082
45%	50%	1921.596	4.505035
65%	25%	2723.239	5.831987

The central diagnostic reaches only 4.505035M gross warrant shares by Q4 2027, versus the 7.945399M full reservation already allowed in the stress envelope. This leaves substantial timing cushion; it does not prove a contractual vesting date. The filing also reports a $1.775M noncash “warrant issued and vested” item without an updated exact cumulative vested-share count. It is an accounting dollar amount, not an additional share count to stack on the reservation. The $2.1M H1 warrant-related reduction of revenue is separate from dilution; gross versus net modeled ASP must be reconciled before adding any further contra-revenue deduction. [AS1, AS4, AS6]

Future grants and funding remain bounded assumptions. A new **1.0M-share allowance over August 20, 2026-Q4 2027**, with 0.5-1.5M sensitivity bounds and 25%/50%/25% subjective weights, is an explicit adequacy check inside the retained 105M total. It is not a disclosed grant or a new amount added to that total. The prior year's 0.87% net-grant dilution multiplied by 74.998M year-end common gives an approximate 0.652M annual reference; extending it over 1.36 years gives roughly 0.890M, rounded to 1.0M for this diagnostic. The proxy's burn-rate definition and a year-end rather than average denominator limit precision. Performance awards can pay 0-200%; accordingly the additional 0.629463M PSU row is a conservative sensitivity because the August prospectus does not expressly label its count as target or maximum. Even the full Amazon reservation, this additional PSU sensitivity and 1.0M future grants sum to **104.77983237M**, leaving **0.22016763M below 105M**. The 2.5M new plan reserve is authority to make awards, not presently outstanding dilution. [AS1, AS2, AS7, AS16, AS17]

That cushion is not unlimited financing capacity. As a separate stress, assign a subjective **30% chance of another $600M gross equity program by Q4 2027 and 70% of no such additional program**; this is a risk diagnostic, not a sourced company plan or an active share addition. At $80/$100/$150 issue prices it would create 7.5M/6.0M/4.0M shares, respectively, and approximately $588M cash at a 2% selling cost. Under the central customer-payment proxy, the disclosed PSU count and

1.0M new grants, the gross diagnostic is 100.710005M; an additional 6M shares would exceed 105M. The combined full-warrant, maximum make-whole and larger-grant stresses can also exceed 105M. Three 2026 ATM programs and substantial factory cash uses justify monitoring these tails, but no complete post-June cash budget supports a precise new financing forecast. If a financing state is adopted later, compute its price target with its own shares and funding consequences before weighting; do not treat the present 105M as an independently calculated probability-weighted mean.

The period-specific share convention is retained. Gross reserved shares are useful to test the Q4 2027 allowance; they are not interchangeable with diluted EPS. GAAP Q2 diluted shares equaled the 81.568M basic weighted average because the company had a loss; non-GAAP diluted shares were 88.152M. The 92.8M Q3 guidance was issued August 6, before this ATM. Without daily sales dates, the offering's Q3 weighted contribution is only bounded from zero to 2.537920M shares; this is not an automatic addition to guidance. The current 92.8M remains a dated forecast pending actual results. [AS1, AS11]

Likewise, Q4 2026's 96M forecast is not a gross all-awards-reserved ceiling: the mechanical common bridge plus ordinary conversion, all August awards and only the initial Amazon tranche totals **96.52920337M**. Actual weighted dilution depends on issuance/vesting dates, stock price, assumed award proceeds, tax withholding, and the applicable treasury-stock or if-converted treatment. Amazon can also elect cashless exercise using the contractual 30-day VWAP; full-reservation cashless delivery would be approximately 6.062689M shares at a $100 VWAP, but the model's target price is not that VWAP. This example is not a switch to a new economic convention. Near-term exact EPS reconciliation therefore remains pending; the forward forecast is retained with the limitation visible.

Quarter	Retained shares M	Price target USD
Q3 2026	92.800000	148.974435
Q4 2026	96.000000	246.892192
Q1 2027	99.000000	349.513946
Q2 2027	101.000000	415.844407
Q3 2027	103.000000	454.403702
Q4 2027	105.000000	463.058351
Q1 2028	106.000000	447.823053
Q2 2028	107.000000	Unavailable: incomplete forward window
Q3 2028	108.000000	Unavailable: incomplete forward window
Q4 2028	109.000000	Unavailable: incomplete forward window

Cash and debt follow the same transactions once. June 30 cash and equivalents were $499.737M, with $9.021M restricted cash; subsequent ATM proceeds are financing cash and cannot be added to June cash to claim an October cash balance. H1 operating cash use was $73.781M; cash PP&E purchases were $335.132M and equipment/other prepayments $289.744M, within total investing cash use of $633.713M. Deposits later applied to delivered equipment must not be counted twice. The September 4 property purchase alone used $26.783472M. Factory leases, working capital and further equipment spending are additional reasons to reconcile the next cash-flow statement. Ordinary conversion provides no second cash inflow; legal debt and its $3.435438M annual coupon disappear only on settlement, with any cash settlement also reducing liquidity. Warrant cash exercise may provide cash, while cashless exercise does not. Neither hypothetical exercise cash nor the ATM proceeds receive an independent net-cash addback under the unchanged OP valuation method. [AS1, AS3, AS5, AS8-AS10]

Conclusion and remaining disclosure limits. The instrument inventory closes the previously unquantified reserve gap and supports retaining **105M / USD 463.058351** without duplicating the ATM. It does not establish an exact October 5 company-wide legal count, actual Q3 diluted EPS, current Amazon vested/exercised count, the redacted purchase thresholds, the PSU target/max basis, future discretionary grants, or future financing. Those specific items replace the old blanket "instrument reconciliation incomplete" statement. A material new financing, award, conversion election or cash-flow shortfall triggers a denominator and funding review; it is not assigned additional revenue, OP or a Base change in advance.

Capital reconciliation sources

- **AS1** Q2 2026 Form 10-Q, filed August 6, 2026. original HTML and extracted text saved.
- **AS2** August 21, 2026 ATM prospectus, common and award counts as of August 20. original HTML and extracted text saved.
- **AS3** October 5, 2026 issuer ATM completion release. original HTML and extracted text saved.
- **AS4** March 13, 2025 Amazon warrant, original executed exhibit. original HTML and extracted text saved.
- **AS5** December 23, 2024 indenture for the 2.75% notes due 2030. original HTML and extracted text saved.
- **AS6** FY2025 Form 10-K, filed February 26, 2026. original HTML and extracted text saved.
- **AS7** April 24, 2026 definitive proxy, equity plans and award statistics. original HTML and extracted text saved.
- **AS8** September 10, 2026 Form 8-K: September 4 property purchase. original HTML and extracted text saved.
- **AS9** September 1, 2026 Form 8-K: August 31 lease agreements. original HTML and extracted text saved.
- **AS10** September 15, 2026 Form 8-K: September 10 Ningbo lease. original HTML and extracted text saved.
- **AS11** August 6, 2026 Q2 results and Q3 guidance. primary page reviewed through web; no local original captured.
- **AS12** October 5, 2026 Form 144, Joshua Yeh. primary page reviewed through web; no local original captured.
- **AS13** September 17, 2026 Form 4, Richard Black. primary page reviewed through web; no local original captured.
- **AS14** September 10, 2026 Form 4, Fred Chang. primary page reviewed through web; no local original captured.
- **AS15** September 10, 2026 Form 4, William Murry. primary page reviewed through web; no local original captured.
- **AS16** June 5, 2026 Form 8-K: June 4 approval of the 2026 equity plan. primary page reviewed through web; no local original captured.
- **AS17** 2026 equity plan Form S-8, 2.5M new shares registered. primary page reviewed through web; no local original captured.

Q4 2027 price target

Current model item	Value
Q4 2027 price target	463.0584 USD/share
Valuation window	Q4 2027 to Q3 2028
Artifact analytical Base	72/100
Protected portfolio Excel Base	72/100

A 12% joint case substitutes a small amount of existing 1.6T production with 3.2T, including a throughput penalty. External ELSFP receives a separate small pricing contribution.

The sanctioned valuation remains **20x forward-four-quarter operating profit**, with the existing accounting basis, ownership, shares and currency conversion. The active price target next rolls on January 1, 2027.

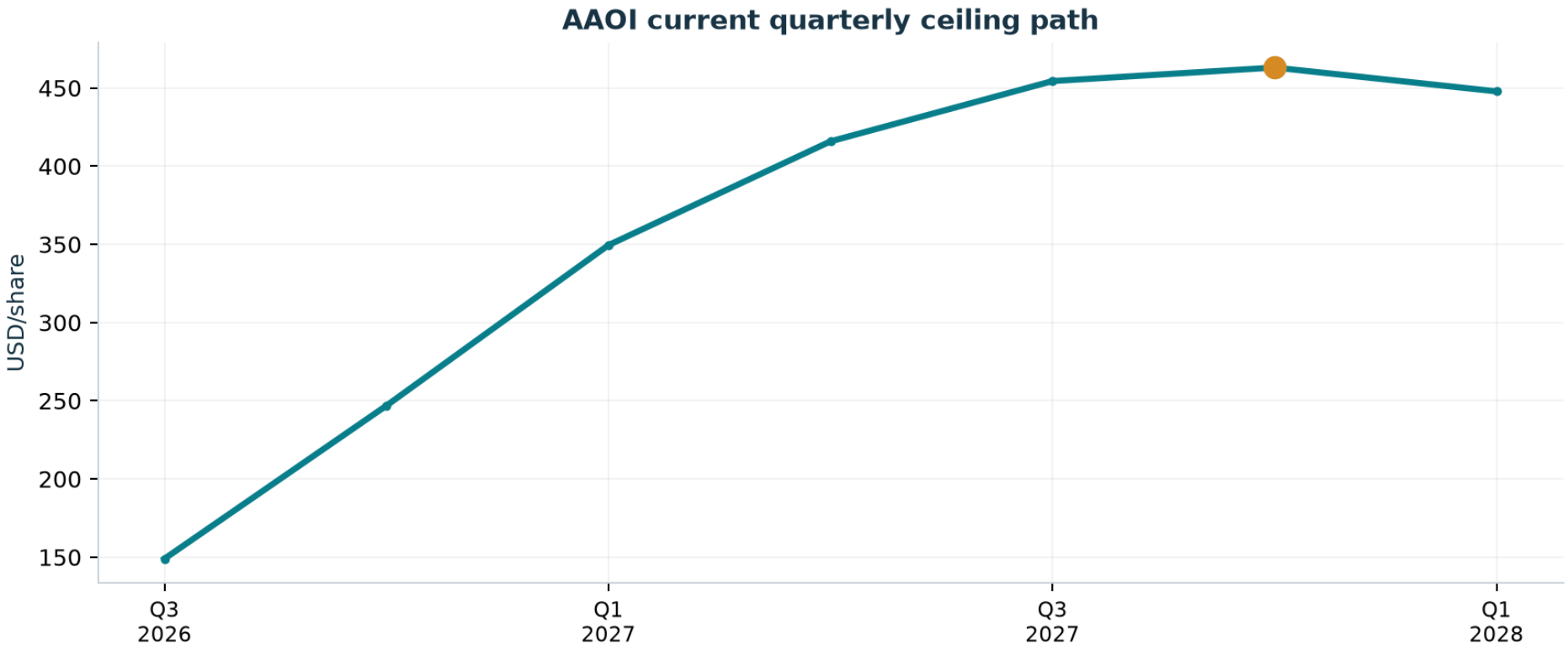


Figure 1. Current quarterly price targets, USD/share. Gold marks the active Q4 2027 horizon. The 40% policy contribution is included once in operating earnings.

price target quarter	Current price target USD/share
Q3 2026	148.9744
Q4 2026	246.8922
Q1 2027	349.5139
Q2 2027	415.8444
Q3 2027	454.4037
Q4 2027	463.0584
Q1 2028	447.8231

Analytical Base and scored conviction

Category	Score	Evidence and limit
Core Thesis	14/20	US manufacturing and internal lasers support execution; standardized modules and competing qualified suppliers limit monopoly status.
Cycle Stage	10/10	800G ramp and 1.6T qualification-to-volume transition with explicit capacity milestones.
TAM Expansion	8/10	Increasing optical bandwidth and distinct external-light-source applications, subject to capture and non-additivity.
Catalysts	5/5	Texas capacity, qualified output, 1.6T mix and CPO customer conversion.
Macro	9/10	AI interconnect investment and high-speed adoption expand the demand pool.
Alpha	2/5	The analytical edge rests on company unit economics and timing rather than broad theme discovery.
Risks	-5	Five deductions detailed in Section 11.
Total	43/60 = 72/100	Watchlist; price-independent analytical thesis score

Artifact analytical Base: 72/100. Protected portfolio Excel Base: 72/100. The portfolio value is a separate retained decision; it is not a scoring-table input.

U.S. manufacturing and internal laser supply are already credited in Core 14. The low-confidence 3.2T slot substitution and filtered ELSFP premium improve modeled economics without yet improving qualification evidence or erasing the five distinct risk classes. Geography is not rewarded twice.

Base measures thesis quality. The 40% policy probability weights only the incremental operating economics, once; neither Base nor the policy weight is applied again to the final price target. Scores use integer category points on the /60 framework lattice.

Policy breadcrumb and economic hypothesis

Serenity's October 1 post attributes Washington meetings to Morgan Stanley and reports a possible 3.2T-focused restriction, with a potential 65% U.S.-value route for Chinese modules. The source expects laser pricing to matter more than broad module-share redistribution. Treat this as an attributed policy scenario; the original MS note and final FCC language establishing these exact terms were not recovered. The separate September 25 Senate proposal concerns sensitive federal systems and a five-year transition. It is not confirmation of these FCC terms.

Serenity policy post | Foundry follow-up | Separate Senate proposal

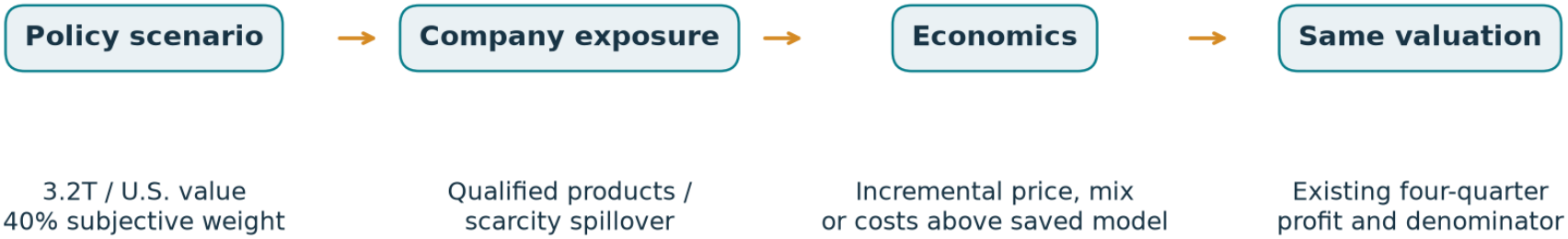


Figure 2. Analytical transmission of the attributed policy scenario. This is a model diagram, not an official regulatory diagram or customer award.

Assigned common policy-scenario probability: 40%. Subjective initial weight above the single-source 25-35% band because Serenity has the owner's highest weighting and attributes Washington meetings to Morgan Stanley. The exact future rule, timing and economic effect remain uncertain. Serenity and Morgan Stanley are one source chain, not two independent confirmations.

3.2T orders are not reported. Conditional on the policy scenario, a 30% conversion judgment weights a bounded allocation of existing 1.6T production slots to 3.2T. GS's existing 2028 3.2T industry ASP is a proxy, not an AAOI contract. Each 3.2T module consumes 1.25 reference 1.6T slots and 1.5x unit COGS, both labeled complexity proxies. No assembly throughput is added. Product yield and line-hours are not disclosed. Joint probability is 12%, not an additional full factory. A separate ELSFP price effect applies only to external ELSFP sales, without using those same units in transceiver revenue.

Eligible fractions are bounded scenario assumptions, not origin certificates. U.S. corporate domicile and a U.S. listing do not prove manufacturing origin. If the final content definition excludes the modeled products, their eligible fraction goes to zero.

For component and integrated suppliers, the filtered revenue pool means 3.2T-linked sales plus incremental cross-generation pricing spillover from competition for common qualified capacity. It is not a claim that 800G/1.6T is directly regulated. Fractions are LOW-confidence exposure proxies, not verified geographic market shares.

Signal ledger field	Current treatment
Signal ID / date	FCC-OPTICS-20261001; Serenity/MS source chain
What is already included	All saved product growth, shortages, capacity, customer probabilities and contracts. The bridge adds only the explicitly calculated incremental economics.

Signal ledger field	Current treatment
Probability and timing	40% common policy scenario; price/cost intensity 0 before Q3 2027, 25% Q3 2027, 50% Q4 2027, 75% Q1 2028, 100% from Q2 2028. AAOI mix uses its own slot ramp.
Numerical contribution	Q4 2027 operating quarter: delta revenue 0.489600, delta consolidated OP 0.440640 (USD millions). Forward-four-quarter valuation OP changes 20.769827.
Dependence	The policy probability is shared across companies. These are correlated portfolio exposures, not independent confirmations. Existing customer-conversion probabilities are already inside source revenue.
Raise weight	Rule/proposal scope supports qualifying content and timing; company product/foundry evidence improves; customer sourcing or like-for-like price negotiations support the incremental premium.
Reduce weight	Policy is delayed beyond the horizon, the content definition excludes products, supply eases, fixed contracts prevent repricing, unit costs absorb the premium, or 3.2T conversion slips.
Zero-contribution boundary	No broad TAM uplift, no new factory capacity, no OCS benefit and no blanket-ban recovery. AAOI receives no modeled units before 2028. MXL is assessed separately as a candidate.

Conditional inputs selected before calculating targets

Input	Value	Evidence status
3.2T conversion conditional on policy	30.00%	LOW-confidence analyst assumption; inherited cap where labeled
2028 3.2T ASP proxy (USD/module)	1,200.00	LOW-confidence analyst assumption; inherited cap where labeled
3.2T / 1.6T unit COGS	1.50	LOW-confidence analyst assumption; inherited cap where labeled
Incremental opex / incremental revenue	5.00%	LOW-confidence analyst assumption; inherited cap where labeled
ELSFP relevant qualifying share	50.00%	LOW-confidence analyst assumption; inherited cap where labeled
ELSFP repricable share	60.00%	LOW-confidence analyst assumption; inherited cap where labeled
ELSFP price premium	5.00%	LOW-confidence analyst assumption; inherited cap where labeled
ELSFP incremental COGS rate	5.00%	LOW-confidence analyst assumption; inherited cap where labeled
1.6T-equivalent slots / 3.2T module	1.25	LOW-confidence analyst assumption; inherited cap where labeled

Incremental COGS and opex rates apply to incremental price revenue, not original product revenue or COGS. AAOI mix cost uses unit COGS separately. AXTI stranded COGS is a fraction of COGS attached to lost units. Chinese component-cost rates apply to affected reference COGS.

Timing is an analyst sensitivity. A deliberately labeled early-sourcing proxy ahead of a 2028 3.2T ramp. Earlier quarters receive zero incremental policy revenue. Full intensity begins Q2 2028 and remains an uncertain continuation through the saved forecast only. These dates are not announced orders or a published regulatory timetable. The probability sensitivity below includes a no-policy control; a later start would reduce the near-term contribution.

3.2T substitution and throughput budget

The 40% policy outcome and 30% conditional AAOI commercialization outcome imply **12% joint weight**. Each new 3.2T module uses **1.25 existing 1.6T-equivalent production slots** and **1.5x 1.6T unit COGS**. These are LOW-confidence process proxies, not disclosed throughput or yield. The conditional allocation uses 5%, 10%, 15% and 15% of the existing 1.6T slot pool across 2028. This is incremental policy-linked conversion above the saved no-3.2T path. It is not all potential AAOI 3.2T demand.

Quarter	Reference 1.6T units M	Displaced slots M, conditional	3.2T units M, conditional	Weighted 3.2T units M	3.2T unit cost USD
Q1 2028	1.659353	0.082968	0.066374	0.007965	536.372
Q2 2028	1.786995	0.178699	0.142960	0.017155	509.571
Q3 2028	1.914638	0.287196	0.229757	0.027571	487.358
Q4 2028	2.042280	0.306342	0.245074	0.029409	466.546

A 3.2T invoice replaces the foregone 1.6T invoice; only net incremental profit reaches the price target. The separate ELSFP price contribution covers external ELSFP sales and is not multiplied by the 30% transceiver-commercialization probability. The generic pricing phase is not applied a second time to the explicit 3.2T slot ramp. The existing forecast ends in Q4 2028, leaving seven complete valuation windows.

Incremental revenue, cost and profit

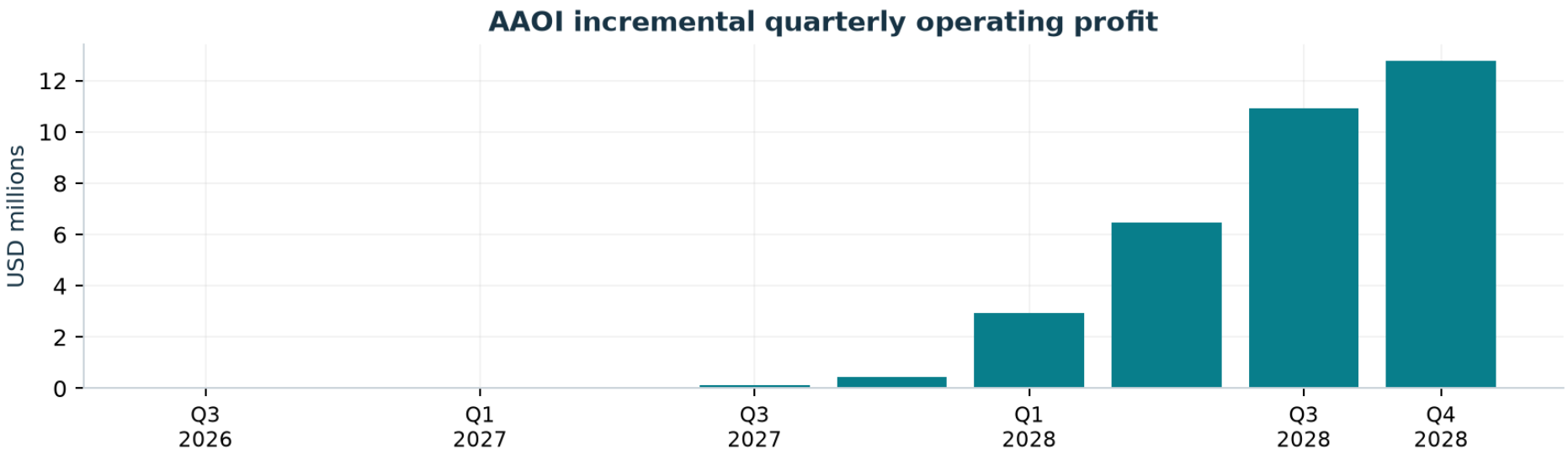


Figure 3. Incremental consolidated operating profit, USD millions. Positive bars show the net probability-weighted policy contribution; baseline operations are unchanged.

Quarter	Delta revenue	Delta COGS	Delta gross profit	Delta opex	Delta OP
Q3 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q4 2026	0.0000	0.0000	0.0000	0.0000	0.0000
Q1 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q2 2027	0.0000	0.0000	0.0000	0.0000	0.0000
Q3 2027	0.1296	0.0065	0.1231	0.0065	0.1166
Q4 2027	0.4896	0.0245	0.4651	0.0245	0.4406
Q1 2028	3.8752	0.7504	3.1248	0.1938	2.9311
Q2 2028	8.4009	1.5100	6.8910	0.4200	6.4709
Q3 2028	13.9165	2.2935	11.6231	0.6958	10.9272
Q4 2028	15.8943	2.3415	13.5528	0.7947	12.7581

Active-window product bridge

Product / event	Pre-policy operating pool	Filtered exposure	Weighted delta revenue	Weighted delta OP
External ELSFP price	163.2000	48.9600	0.4896	0.4406

Filtered exposure is a scenario allocation, not verified U.S.-origin revenue. For cost cases the exposure is an affected cost pool; AAOI substitution units are explained in its separate bridge. All financial amounts use the company table units.

Business research and source evidence

The following sets out the company thesis, historical results, guidance, capacity, program probabilities, risks and original source images. Source dates remain visible. Operating assumptions below feed the complete current financial tables, including the explicit probability-weighted policy contribution.

Inherited breadcrumb inputs and source images

These probabilities are explicit investment judgments under Framework 8.0, not measured statistical frequencies or claims of confirmed orders. Unconfirmed but credible evidence can affect the model when its company-specific transmission is quantified.

Higher800G/1.6T factory utilization

Inherited probability: 30% extra five-point utilization case.

Model demand pulling more output through the existing high-speed capacity. Retain the current lower ASP path, shared capacity allocation and 1.6T timing.

Operating bridge: Conditional 800G and 1.6T utilization rises 5 percentage points from Q1 2027, capped at 95%. Apply probability 30% only to incremental units and their existing product contribution margins.

Limits and counter-evidence: Both speed lines retain the same allocated capacity: no double use of the factory. DSP/laser shortages can prevent this uplift. Annual GS TAM remains unchanged. Model ends Q4 2028; later price targets stay unavailable.

Optic takeaways from APAC Conf: debates on supply and market-share dynamics

Industry Overview

Global 800G/1.6T demand could reach 200mn units by '27

During Sep 17-23, we held investor meetings with optical communications suppliers, including Innolight, CIG, DS Precision, and Everbright, as well as upstream suppliers like Crystal-Optech/Sunny Optical/Himax. Supply chain indicated solid transceiver demand through 2027, with consensus expectations for both 800G/1.6T demand to reach 90-100mn units each. However, shipments will likely remain constrained by upstream component supply, with DSP the most cited bottleneck, particularly for 1.6T. mSAP PCB and CW lasers are also in tight supply. Although investor feedback suggests less concern about 800G shipments, Innolight noted that both 800G/1.6T could face supply shortages.

Tier-2 suppliers offer positive guidance

Tier-2 suppliers are providing positive guidance for 2027: (1) DS Precision guides to 35mn units, including 10mn of 1.6T; (2) CIG targets monthly output of 2mn units in 2027 vs 1.5mn by end-2026, implying potential annual output of 18-24mn units, with more than 40% from 1.6T; and (3) Luxshare expects output of 6-8mn units in 2027, with 1.6T accounting for c.50%. All three highlighted sufficient component supply for 2027 and

User-supplied evidence for Higher800G/1.6T factory utilization. The quoted forecast or estimate retains the source status described above. Original exhibit, panel 1 of 2; read from top to bottom.

said they had secured DSPs for 1.6T transceivers. However, investor feedback indicates greater confidence in tier-1 suppliers' positioning and a preference to discount tier-2 vendors' guidance because of concerns about DSP supply. Our checks suggest potential DSP volume of 60-70mn units in 2027, broadly in line with most investors' views.

NPO offers potential upside into 2028

All transceiver makers indicated they are working on 3.2T/6.4T NPO projects. Innolight noted that, in a bull case, the market could reach 30-50mn OE units in 2028, vs current expectations of 2-5mn units in 2027 and more than 20mn in 2028. Nvidia's NPO project remains in the sampling stage, and more concrete orders could emerge toward year-end or in early 2027. On design, we see both integrated-laser and ELS approaches, while the supply chain is seeking to use more 100-150mw CW lasers even in ELS solutions.

Passive: more smartphone optic suppliers enter market

More smartphone optical suppliers are expanding into passive components, initially focusing on FAU-related for transceivers with a goal of entering CPO. Crystal-Optech noted that smartphone optical suppliers benefit from greater coating-equipment capacity as equipment lead times have lengthened. However, contributions are unlikely before 2027, as testing/certification typically take a year. Sunny said it is in sampling stage and engaging more closely with domestic customers. On CPO, Himax highlighted its close partnership with FOCI and is working on a 3rd-gen design. Himax noted that 2027 will remain an engineering-run stage but looks positive on future growth.

User-supplied evidence for Higher800G/1.6T factory utilization. The quoted forecast or estimate retains the source status described above. Original exhibit, panel 2 of 2; read from top to bottom.

Sources: APAC.

Transmission field	Inherited treatment
Probability	30% extra five-point utilization case
Recognition	Conditional 800G and 1.6T utilization rises 5 percentage points from Q1 2027, capped at 95%. Apply probability 30% only to incremental units and their existing product contribution margins.
Overlap / uncertainty	Both speed lines retain the same allocated capacity: no double use of the factory. DSP/laser shortages can prevent this uplift. Annual GS TAM remains unchanged. Model ends Q4 2028; later price targets stay unavailable.
Raise probability	Timely qualifying milestones, comparable revenue scope, stronger customer allocation and observed unit economics.
Lower probability	Delay, customer substitution, weaker realized economics, overlap or evidence contradicting the stated source scope.

1. Investment thesis and supercycle position

AAOI provides an investable exposure to the shift toward higher-speed optical modules through its US manufacturing footprint, internal laser capability, and flexible 800G/1.6T assembly lines. The investment thesis requires qualified output and a richer product mix to convert a large market into company revenue. It is a temporal manufacturing and qualification bottleneck, rather than a demonstrated monopoly in standardized transceivers.

The operative S-curve is an 800G ramp followed by a faster 1.6T mix shift. External light-source modules for CPO provide another revenue vector, subject to customer qualification, high-power laser capacity and packaging yield. A laser consumed inside an AAOI transceiver is an input cost; it is not a second external revenue sale.

The model pools high-speed module capacity, reaches 91.5% probability-weighted saleable utilization in Q4 2027, and shifts 1.6T to 60% of high-speed units at that point and 80% by Q4 2028. It holds total high-speed capacity at 930,000 units per month during 2028. That is a finite modeling assumption, not a company statement that expansion stops. Pricing declines eventually outrun mix improvement at that capacity level.

The 3.2T industry ramp belongs in the active research horizon. AAOI-specific qualification, orders, testing throughput and a production schedule are not established in the materials reviewed, the reference had no 3.2T revenue. The current model assigns a 12% joint probability to a bounded 2028 substitution of existing 1.6T slots, with explicit unit-cost and throughput proxies. Higher market volume alone does not grant AAOI a constant market share or remove component bottlenecks.

TAM expansion lens

Phase	Application and mechanism	Framework treatment	Included company revenue
Phase 1	AI training/inference clusters and hyperscale interconnects; more deployed optical links and greater bandwidth per link	100% thematic weight; company execution still requires evidence	Modeled 800G/1.6T modules and the separately identified ELSFP vector
Phase 2	Wider cloud, edge and physical-AI deployments requiring high-speed optical connectivity	50% thematic weight; assess addressable products and customers	No separately added wallet; avoid overlap with datacenter forecasts
Phase 3	Further distributed-compute architectures and new photonics applications	20% thematic weight; no assumed design wins	Unmodeled

These framework phase weights are research conventions, not probabilities applied to the revenue table. They do not justify adding the same optical demand twice. AAOI's expansion score reflects a widening application set tempered by competition and the need to qualify each product generation.

Engineering: qualification has several separate gates

The March 1 Irrational Analysis article is useful for understanding why an optical module can pass device reliability testing and still wait for customer acceptance. Its reported sequence is reliability and manufacturing-line qualification, followed by interoperability work across the customer's switching platforms. The author proposes a transmitter-equalization coefficient mapping problem as the remaining firmware issue. That is a technically plausible explanation, not an AAOI-confirmed root cause. A subsequent shipment does not retrospectively prove which register, vendor or training algorithm caused the delay. [E-AAOI1]

Michael Edward Genovese: Great. Thanks very much. So it sounds like instead of having a steady kind of ramp, this 800G we are expecting to come in with really big numbers starting 2Q this year and then, you know, huge numbers next year—by 2Q next year. I guess for the ramp in 2Q this year, could you just go over some of the milestones, maybe talk about the issue on the firmware, but also the ongoing qualification milestones that you have to hit to kind of have that ramp in 2Q?

Stefan Murry: Right. So as you mentioned, in order for the ramp to start in earnest, our 800G products have to be interoperable with all the different platforms that are out there at this particular customer. There is a lot of them. So the firmware has to be modified to work with all those different platforms. Hardware-wise, everything is fine. No problem with that. Firmware is good on most of the platforms. We just have to make some tweaks to get it to work across all of the different platforms that they have. And that is really—the customer and us have agreed that should be done in the middle of next month.

So a couple of weeks, three weeks from now, something like that. And then that is basically the last hurdle to kind of unleash the ramp. As we talked about, we have already started manufacturing products for that ramp. So from a manufacturing standpoint, we are gearing up. Just to touch on the beginning part of that question, too, about kind of nonlinearity of the ramp, that is because what you are seeing is our production capacity coming online. Right? It is not gated by demand, it is gated by our ability to produce, and that does not come on in a linear fashion. Right? You build a production line and you get a step function, not a smooth ramp.

Thompson Lin: Yes. Let me emphasize. Because every customer has so many different switch, transceiver supplier, so many different kind of ASIC. And, you know, when a customer adds more switch, this will change the firmware. We were supposed to get a green light to ship out already in December. The delay is not our problem or whatever. It is because, you know, how come build AI—you know, the whole system is much more complicated than before. That is why it takes much longer. And right now, I think we feel very comfortable. And right now, let me say, we have got almost two years of loading forecast from more than one customer. Let me say that, okay? For 800G.

Engineering exhibit aaoi-I07. Saved source: AAOI Probably Worth Chasing, 2026-03-01. Attribution and scope are discussed below.

Historical management-call excerpt reproduced in the March 1 article. Firmware and platform interoperability were management's stated focus at the time. The precise TxFIR diagnosis belongs to the author; it is not a company disclosure.

Equalization compensates for an electrical channel's loss and distortion. The article's coefficient examples explain why host/module settings and firmware can prevent a working optical design from passing a particular customer's tests. Its IEEE 802.3ck tables concern electrical interfaces based on 100 Gb/s signaling; they are not a complete 800G optical-module specification. The illustrated seven-bit register mapping is a teaching example, not an AAOI register dump or measured failure. The correct research question is whether the specific customer platform has passed interoperability and acceptance at the required error rate, temperature and operating conditions. [E-AAOI1, E-AAOI3]

The historical uncertainty has a dated outcome: AAOI's May 7 release reports its first volume 800G shipment to a large hyperscaler during Q1 2026. Its March 23 order release had also described more than $53M of 800G orders for delivery in Q2 through mid-Q3 following qualification. Those events precede the current Q2 actuals and Q3/Q4 ramp assumptions. They support the existing production path; they are not newly available capacity on October 6. [E-AAOI2, E-AAOI4]

Gate	Economic transmission	What would change the current model
Laser/device reliability	Burn-in, lifetime and thermal failures can raise scrap, screening cost and field returns	Product-specific qualified failure data and associated cost or shipment impact
Manufacturing line	Installed tools become repeatable good output	Accepted yield, test time and qualified monthly shipments, rather than floor area
Host interoperability	A good module may still fail link training or firmware compatibility on one platform	Dated customer/platform acceptance or a new quantified delay
Customer acceptance and delivery	Finished inventory becomes recognized sales	Actual shipped units, realized ASP and acceptance terms

Probability and overlap decision. The historical first volume shipment is a reported fact, recorded at 100% event status. The model retains the separate **30% probability of an extra five percentage points of 800G/1.6T utilization from Q1 2027, capped at 95%**, with existing ASPs, shared capacity and incremental contribution margins. The old firmware-resolution narrative overlaps that already incorporated commercial ramp completely; incremental units, revenue, COGS, opex and OP from this article are all zero. This is a date and overlap decision, not an assumption that unconfirmed engineering signals have no value. No additional Base point is awarded for the same qualification progress.

For a future delay, the bounded exposure is the affected customer's product and acceptance quarter. Revenue at risk equals delayed accepted units times that product's ASP; OP impact must deduct avoidable production costs while retaining idle-factory, rework and customer-support expense. Moving all company revenue, or treating delayed units as both lost forever and reshipped later, would overstate the effect. ELSFP remains a separate qualification problem: successful 800G firmware work does not validate high-power laser yield or the model's earlier ELSFP capacity timing.

Engineering sources, reviewed October 6, 2026: E-AAOI1: Irrational Analysis, March 1, saved full body and all 16 image elements reviewed; E-AAOI2: AAOI Q1 results, May 7; E-AAOI3: IEEE 802.3ck task force and scope; E-AAOI4: AAOI March 23 order release.

2. Actual financial anchor and company guidance

The actual anchor is Q2 2026. Company figures below are distinguished from the quarterly model. [S1, S2]

Q2 2026 actual metric	Value	
Revenue		$191.922M
Data center		$107.662M
CATV		$80.578M
Other revenue		$3.682M
Non-GAAP gross profit		$57.253M
Non-GAAP gross margin		29.8%
Non-GAAP operating expenses	Approximately $67.6M	
Non-GAAP operating loss	Approximately $(10.3)M	
Non-GAAP EPS		$0.06
GAAP net loss	Approximately $(22.8)M	

Positive adjusted EPS in that quarter does not establish positive operating profit. The valuation below uses modeled operating profit, not EPS, subsidy income or tax benefits.

Formal Q3 2026 guidance: revenue $255-290M, non-GAAP gross margin 29-30.5%, and approximately 92.8M diluted shares. [S1] The management call supplies additional operating-expense, product-ramp and capacity context. Those statements are dated management expectations rather than reported subsequent results. [S2]

The prior company artifact records management's expectation of more than $500M revenue in Q4 2026, a $70-80M 1.6T contribution, Q1 2027 1.6T revenue at least doubling, and a mid-2027 datacenter-transceiver milestone of roughly $471M monthly. It also records a $200M 1.6T purchase order and customer demand above production capacity. These are company-specific demand anchors, not additional amounts to add to the quarterly revenue schedule. [S2, S6]

The model's Q4 2026 gross margin is 33.98%, above the 32-33% exit-margin comment recorded from the call. This is an explicit model assumption, including the five-point 1.6T margin premium; it must not be described as matching that margin comment. The artifact carries this discrepancy as a validation item rather than representing all modeled financial metrics as guided.

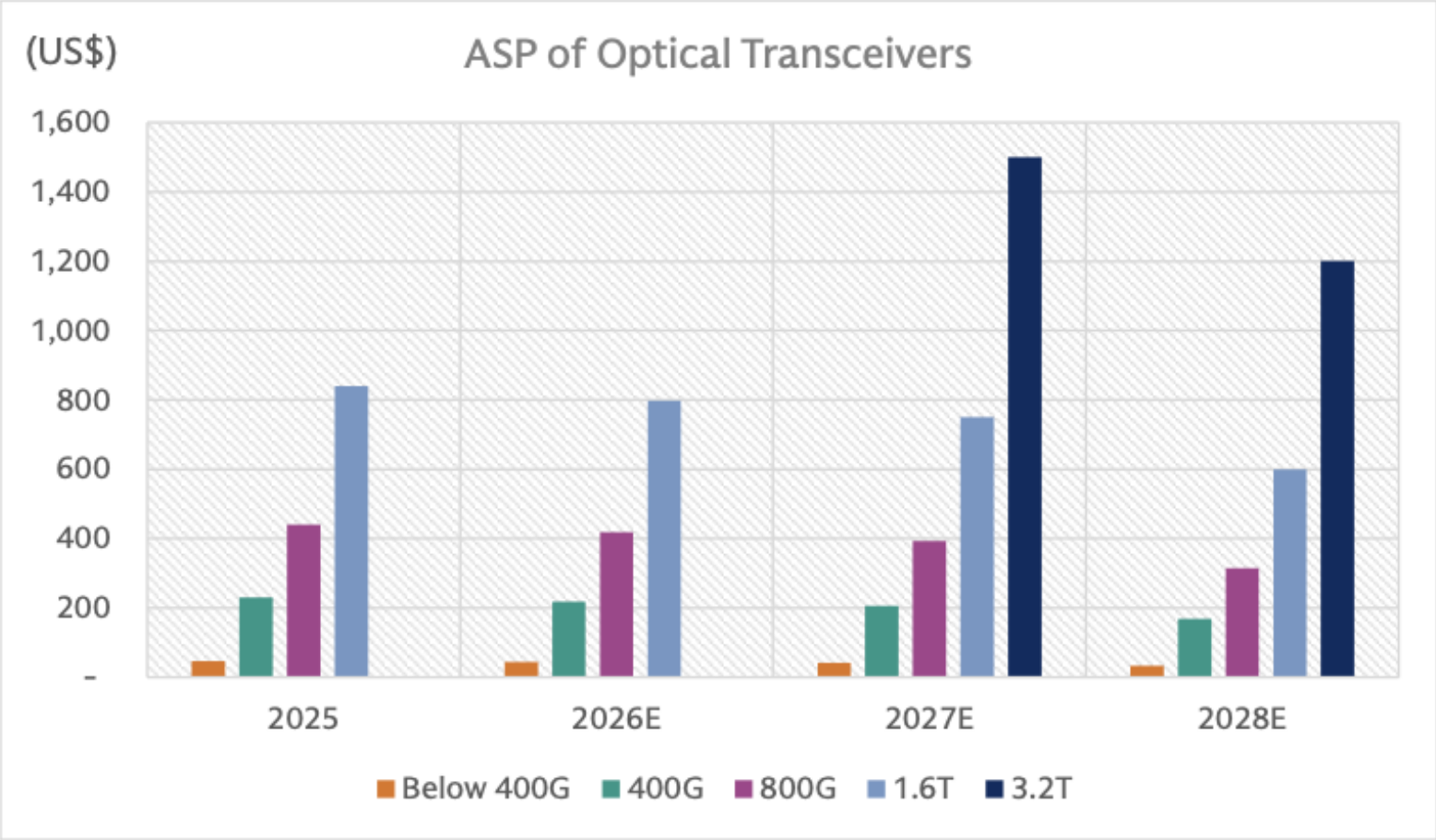
Company capacity milestone	High-speed modules per month	Treatment
End Q1 2026	About 100,000	Reported historical capacity context
At Q2 2026 call	Approaching 200,000	Dated management statement
End 2026	Around / over 650,000	Management forecast endpoint
End 2027	Over 930,000	Management forecast endpoint
During 2028	No incremental high-speed capacity established in this model	Model holds 930,000; expansion remains an open research input

Management describes 800G and 1.6T as sharing manufacturing lines, with different final testing requirements. The model uses that flexibility to change product mix. Pooling nominal assembly capacity does not prove interchangeable final-test throughput, sufficient DSP/TIA availability, laser capacity, or customer qualification. [S2]

The inherited company material describes Texas capacity additions, including initial production at the 210,000-square-foot facility in late Q3 2026 and additional facilities expected in early 2027. Monitor actual production acceptance rather than treating construction area as shipped units. The memory-related switch shortage can also suppress legacy 100G shipments even while high-speed optics demand rises. [S2, S6]

3. Optical market baseline and company capture

Exhibit 4: Global Optical transceivers: Assuming ASP decline YoY for like-for-like products



Source: Company data, Goldman Sachs Global Investment Research

Supplied GS Exhibit 4. Module ASP benchmarks; quarterly AAOI pricing retains the original interpolation.

Use Framework 8.0 DB1 for market definitions and source status. The global module/transceiver forecast reported for September 2026 is **$67.7B in 2026, $131.4B in 2027 and $148.5B in 2028**. It is not a company revenue forecast or an additive sum of modules, lasers, substrates and CPO engines. [S3, S4]

The public GS summary reports approximately 49M 800G, 71M 1.6T and 23M 3.2T units in 2027. Its 61% increase to the 2027 1.6T-and-above shipment forecast is a revision to the broker's earlier forecast, not a growth factor to multiply into AAOI's units. [S3]

Approximate annual module ASP	2026E	2027E	2028E
800G		$420	$400	$320
1.6T		$800	$750	$600
3.2T	Not shown		$1,500	$1,200

ASPs are rounded visual readings of the user-supplied Goldman Exhibit 4. They are broad industry benchmarks, not disclosed AAOI realized prices or a like-for-like match for every reach, architecture, customer and manufacturing location. The retained quarterly price path was calibrated to the annual benchmarks using the reference shipment weights and adds no general US-production premium. The separate current ELSFP premium applies to its filtered exposure only. Any premium requires a SKU/customer/contract bridge. [S4]

The reported Rubin Ultra assumptions are two 1.6T and five 3.2T modules per GPU; these are broker-model assumptions, not a verified AAOI order or NVIDIA production BOM. The SiPh percentages reported in the GS summary refer to speed categories: 60% for 800G, 80% for 1.6T and 80% for 3.2T. They do not automatically determine AAOI's architecture mix or laser requirements. [S3, S4]

The supplied CPO shipment revisions are -33% for 2026, +35% for 2027 and +105% for 2028, each relative to the preceding broker shipment forecast. No absolute CPO dollar market or AAOI ELSFP growth rate is inferred from those percentages. ELSFP uses its own capacity, conversion and ASP schedule below. [S4]

2027 company capture cross-check: 4.128M 800G and 3.753M 1.6T units. These are model output requirements, not awarded market shares. Global demand forecasts do not eliminate component constraints.

4. Base score and risk deductions

The analytical score is 43/60, rounded to 72%, following an explicit review of the policy evidence. U.S. manufacturing was already credited; it is not scored again. All portfolio Excel Base values remain protected.

The current summing score is shown in the analytical Base chapter above.

The Base score is a thesis-quality measure. It does not independently validate the modeled utilization, margins or optionality, and no additional Base multiplier is applied to the operating-profit price target.

5. Eight-vector model and source hierarchy

All ten forecast quarters carry **Model** status, including quarters constrained by management guidance. Actual results retain a separate anchor. Product-level forecasts are not upgraded to HIGH confidence simply because an aggregate model fits a guided range.

Revenue vector	Revenue derivation	Evidence / source	Forecast confidence
10G/40G	Explicit quarterly revenue schedule	S6 model assumption; Q2 company product context	LOW
100G	Capacity K/month × 3 × ASP × conversion / 1,000	S6 inherited unit schedule; S2 legacy-demand context	LOW
200G/400G + AOC	Combined capacity K/month × 3 × ASP × conversion / 1,000	S6 mixed-product schedule; no supported SKU split for repricing solely from 400G chart	LOW
800G	Pooled high-speed capacity × utilization × 800G mix × ASP	S2 company capacity/flexibility; S4 price benchmark; explicit modeled utilization/mix	LOW at product level; near-term total revenue is guidance-constrained
1.6T	Pooled high-speed capacity × utilization × 1.6T mix × ASP	S2 company orders/qualification; S4 price benchmark; explicit modeled utilization/mix	LOW at product level; near-term total revenue is guidance-constrained
ELSFP	Independent capacity K/month × 3 × $400 ASP × conversion / 1,000	S2/S6 capacity and module context; modeled recognition/qualified-output conversion	LOW
CATV	Explicit quarterly revenue schedule	S2 near-term demand context; S6 modeled later run rate	MEDIUM near term; LOW thereafter
Other	Explicit quarterly revenue schedule	S6 residual-business assumption	LOW

The company-level proxy is capacity × qualified saleable output × realized price, supported by company demand milestones and checked against the global speed-specific market. A customer-capex-to-AAOI procurement allocation is not available at sufficient granularity to establish individual customer shares. No such shares are fabricated. Forecast confidence remains limited by that gap.

Quarterly revenue derivation

Each high-speed line uses its allocated part of pooled factory capacity x 3 months x expected saleable utilization x quarterly ASP. The 30% scenario adds only the extra output from five points of utilization, capped at 95%. Line mix and capacity are unchanged. Other vectors retain their existing assumptions.

The quarterly ASP path is retained. The annual GS price-shape normalization uses the original shipment weights; changed utilization can make active sales-weighted annual ASP differ slightly from the original benchmark. This avoids repricing the model unintentionally.

Operating expense grows at each quarter's retained reference opex/revenue ratio on the incremental sales. This preserves approved incremental contribution margins.

Guidance and market calibration

Q3 2026 modeled revenue of $270.64M and blended gross margin of 30.33% fall inside formal guidance. Modeled operating expenses of $74.48M fall inside the range recorded in the management call. This anchors the near-term model but does not validate 2027-2028 product ASPs, conversion or costs. [S1, S2]

Q4 2026 revenue is $506.74M and 1.6T revenue is $75.62M. The model's 33.98% blended gross margin is an explicit assumption above the recorded 32-33% exit-margin comment. Q1 2027 1.6T revenue is $321.16M; that projection reflects the active mix/volume assumptions rather than a specific management dollar guide.

Management’s mid-2027 monthly milestones are retained below as dated source evidence. A run-rate milestone and a quarterly average are not identical periods; the current financial chapter supplies the new model comparison.

Monthly transceiver revenue $M	Management mid-2027 milestone
100G/400G	90.00
800G	217.00
1.6T	164.00
Total, excluding ELSFP/CATV/other	471.00

The product allocation requires separate checks. The 1.6T-heavy mix requires additional qualification, components and final-test capacity. Similar aggregate totals do not prove line-by-line convergence; no artificial product plug is inserted to force agreement.

Rolling forward-four-quarter price target

The price target uses operating profit in the stated quarter plus the next three quarters. The company-specific method remains **20x modeled operating profit**. USD millions / period-specific diluted shares in millions. Only seven complete valuation windows; no extrapolated 2029 earnings. Probability is applied to the specified operating increment once; the price target receives no second probability or Base multiplier.

Annual cross-check and demand scale

Calendar buckets sum the stated quarter labels; fiscal proxy mappings remain as in the company earnings section. Annual totals do not replace rolling valuation windows.

Operating driver and probability transmission

No active 2029-2030 operating projection is introduced. Industry demand does not establish additional AAOI capacity beyond the retained schedule. Whole-company revenue including CATV and ELSFP is not divided by a module-only TAM.

9. 3.2T, CPO and optionality ledger

3.2T research gate: the GS industry forecast places this generation in 2027-2028. The reviewed AAOI product overview identifies 800G, 1.6T and CPO/NPO offerings, but it does not establish a company-specific 3.2T volume ramp. Required inputs are a named product/customer qualification, recognized order or allocation, production start, final-test capacity, unit throughput, ASP, margin and displacement of existing output. [S3, S5]

At approximate annual industry ASPs, 100,000 genuinely additional 3.2T modules in a quarter would represent $150M revenue in 2027 or $120M in 2028. If they instead replace 1.6T units one-for-one, the illustrative net revenue increment is $75M or $60M at the annual benchmarks. These arithmetic cross-checks contain no assumed AAOI order, ramp date, throughput equivalence or added profit. No amount from them is included in the current quarterly table.

CPO and high-power laser engineering: external-light-source performance, wavelength control, laser die area, production yield, reliability qualification and packaging yield must be assessed separately. The earlier company material records 300-400mW development and DWDM yield challenges; it does not validate every shipment in the forward ELSFP schedule. Management's component supply position must also be distinguished from DSP/TIA availability and final assembly. [S2, S6]

Item	Current evidence treatment	Revenue treatment	Required next evidence
800G/1.6T company ramp	Company demand/capacity anchors plus explicit model utilization and mix	Reference capacity/mix plus the 30% incremental utilization case	Recognized output, qualification, pricing and product costs
ELSFP	Company product/capacity context; LOW-confidence timing and conversion assumptions	Included once as external module sales	Qualified high-power output, customer acceptance, unit yield and recognized sales
AAOI 3.2T	Company-specific ramp remains LOW confidence	Modeled: 40% policy x 30% conditional conversion = 12%; slot and cost limits in the financial chapter	Product/customer qualification, orders and production schedule
High-speed capacity beyond 930K/month	Not established in the active schedule	Unmodeled	Company expansion timeline, qualified equipment and financing
AAOI pricing premium	A filtered external ELSFP premium is a current assumption, not a quote	5% conditional premium on 50% qualifying x 60% repricable ELSFP exposure; 40% weight	Contract or realized SKU/customer pricing
Broader physical-AI and edge markets	Thematic research opportunity	No separately added revenue	Addressable product and non-overlapping demand bridge

Base and modeled conversion have distinct roles. Conversion represents saleable/recognized output; it is not a hidden probability discount. The accepted probability-weighted utilization increment is inside the existing 800G/1.6T vectors; the policy model separately identifies the 3.2T substitution vector and subtracts displaced 1.6T sales.

10. Competitive position and cross-book implications

AAOI's US footprint and internal laser capabilities may help delivery, customer qualification and sourcing diversification. They do not by themselves establish a durable module monopoly or a guaranteed ASP premium. Competitive pressure must be evaluated by product reach, architecture, customer allocation, quality and manufacturing cost.

An LTA needs a review of price formulas, minimum purchases, allocation rights, resets and cancellation terms. Declining an LTA does not prove prices will rise or that all competitor contracts contain both a price floor and price cap. A signed agreement is not automatically a thesis break; its economics control.

Paired exposure	Shared variable	What to check independently
LITE / COHR / SIVE	High-speed optics and external-light-source demand	Product overlap, qualified laser output, optical power and external sales; avoid treating all laser designs as interchangeable
MRVL / SMTC	DSP, TIA and related component availability	Whether components support AAOI's planned product mix and shipment schedule
AXTI / IQE and substrate/epi suppliers	InP and compound-semiconductor production	Qualified input supply, yield and material intensity; module counts are not wafer starts
TSEM / GFS	SiPh adoption	AAOI-specific architecture and foundry/customer sourcing, rather than industry penetration alone
MU / SNDK / SK hynix / Samsung	Memory constraints and datacenter deployment	Separate positive memory pricing from switch shortages that can delay legacy transceiver demand

These links identify research dependencies. They do not transfer AAOI's revenue forecast, margin assumptions or risk score to another company.

11. Risks and falsification tests

The five risk classes below correspond to the five one-point deductions in the Base table.

1. **Execution and component supply (-1).** The projection requires new capacity to become qualified output, utilization to rise, and product allocation to shift toward 1.6T. Final testing, DSP/TIA supply and laser production can constrain output despite nominal assembly capacity. Counter: material production delay or persistent failure to convert installed capacity into recognized shipments. A Q4 2026 revenue result below $400M would trigger a full thesis review; this is a research threshold, not an instruction to trade.
2. **Competition, concentration and price/cost economics (-1).** Customer concentration and standardized module competition limit pricing power. The GS benchmarks may differ from AAOI's SKU mix; preserving projected margins requires the displayed cost curve. Counter: realized ASP, yield or cost evidence that invalidates the modeled product margins, or customer allocation loss. High-speed price decreases without corresponding cost improvement directly undermine the profit model.
3. **ELSFP performance, yield and timing (-1).** High-power optical performance, wavelength control, external-module qualification and packaging yield may delay recognition. The model assumes nominal capacity reaches 400K/month by Q4 2027, earlier than the public 2028 wording, and a 38% conversion rate is not a disclosed absolute yield. Counter: an announced delay beyond 2028 or no recognized ELSFP revenue by Q4 2027; revise the vector and all affected four-quarter windows.
4. **Funding and dilution (-1).** Factory expansion and working capital can require equity or other financing. Share counts in the price target path are assumptions after the near-term company guide; operating profit is not free cash flow. Counter: share issuance materially above the period-specific schedule or a funding plan that changes capacity timing. Recompute price targets using the actual diluted-share bridge.
5. **Legacy demand and architecture transition (-1).** Memory-related switch constraints can suppress 100G demand, while a faster architecture transition can displace 800G before AAOI captures the next product generation. Counter: persistent legacy weakness, lost design wins or customer adoption of a product AAOI cannot supply on the modeled schedule.

No sixth risk point is added for the same economics under another label. Supplier agreements, LTAs and geopolitical sourcing developments feed the relevant class above. The current category review supports the explicit 43/60 total.

12. Dated catalysts and counters

Window	Required development / positive evidence	Counter / model response
Q3 2026 results	Revenue within the company guide; confirm 1.6T qualification and recognized shipments	Below-guidance results or qualification delay: replace estimates with actuals and rebuild the ramp
Q4 2026 results	Shipment step toward modeled $506.74M revenue and $75.62M 1.6T sales	Reconcile actual product margins; do not claim modeled 33.98% GM was company guidance
H1 2027	Installed equipment becomes qualified output; demonstrate common-line mix flexibility	Testing, DSP/TIA or laser bottlenecks prevent the utilization/mix progression
Mid-2027 / Q3 2027	Track progress toward management's monthly transceiver milestone and test the modeled product split	Aggregate revenue alone does not validate 1.6T share, ASPs or margins
Q4 2027	930K/month high-speed modeled capacity, 91.5% probability-weighted utilization and 60% 1.6T mix; evidence for ELSFP scale	Capacity exists but is not saleable, or ELSFP timing fails the model assumption
2027-2028	Named AAOI 3.2T qualification, order and manufacturing evidence	Reassess the 12% Modeled joint probability, throughput and cost proxies against company-specific progress
Every 2028 quarter	Validate realized ASP and unit-cost declines; assess incremental expansion	Capacity stays fixed while pricing erodes faster than mix improves
Each filing / financing update	Reconcile diluted shares and cash required to fund output	Share count or funding needs exceed the stated schedule

The escalation path is named evidence → qualified design/product → customer order/allocation → recognized shipment. Source credibility can raise research priority before revenue is booked, but timing and unit economics must enter the quarterly model explicitly. Refresh the applicable four-quarter valuation windows whenever actuals, guidance, costs, capacity or shares change.

13. Source register and confidence limits

- **S1 | Company actuals and formal guide:** Applied Optoelectronics Q2 2026 results, August 6, 2026; checked September 20. Primary company release.
- **S2 | Management statements:** AAOI Q2 2026 management call transcript, August 6, 2026. Use attributed management statements; a third-party transcript can contain transcription errors. Company webcast is linked from its events page.
- **S3 | Broker forecast summary:** Public summary of Goldman Sachs's September 7, 2026 optical-transceiver report, checked September 20. The original full Goldman research note was not inspected. This is secondary-source corroboration of the broker forecast, not an independent demand signal.
- **S4 | User-supplied market evidence:** Goldman Exhibit 4 ASP chart and the September 8 market / Rubin Ultra post and ASP / CPO revision post. The chart readings are approximate. The source image is delivered as framework_sources/Goldman_Exhibit4_Optical_ASP_user_supplied.png. Framework 8.0 DB1 defines the scope and limitations.
- **S5 | Company product scope:** AAOI optical-transceiver products, checked September 20. Reviewed materials did not establish a dated AAOI 3.2T volume-production schedule.

- **S6 | User's existing model and artifact:** v3_2_master_portfolio.xlsx, AAOI_Q_Model, and the supplied full AAOI artifact. These supply inherited lower-speed and ELSFP unit schedules, product-margin assumptions, dollar opex and diluted-share assumptions. The source workbook SHA-256 is 387723f831b491f22c17656dd182c3c1a9f7c6697440c2b066f67ca682a3f620. Company-slide values cited inside that model are inherited where the original presentation was not independently retrieved.
- **S7 | Active model decision:** the September 20 user discussion of the GS-price / accelerated-ramp projections and instruction to update the full artifact. The unit mix, utilization, quarterly price interpolation, margin/cost requirements and finite horizon in this document are explicit analyst assumptions authorized for incorporation; authorization does not turn them into disclosed company facts.

Open evidence requirements: AAOI realized ASP by comparable SKU/customer; test-limited 1.6T allocation on shared lines; unit-cost and margin bridge; ELSFP timing/conversion and recognized revenue; additional high-speed capacity after the disclosed milestones; 3.2T orders and qualification; actual Q3 weighted dilution, an updated company-wide legal count, Amazon purchase/vesting detail and future funding against the reconciled total-share allowance. These gaps limit confidence and remain visible in the model rather than being filled with invented inputs.

Current-state controls

This is the current production analysis. The financial model includes the probability-weighted policy contribution; all portfolio Excel Base values remain protected.

Model reconciliation and limitations

The current financial chapter supplies the new forecast tables. Excel Base overrides, valuation methods, share schedules and units remain protected. The next 12-16 months deserve the highest forecast weight; outer-year estimates are substantially less reliable.

Revisit probabilities and operating inputs as evidence changes. Official confirmation is not required to value a quantified breadcrumb; source uncertainty remains explicit.

Complete current operating model

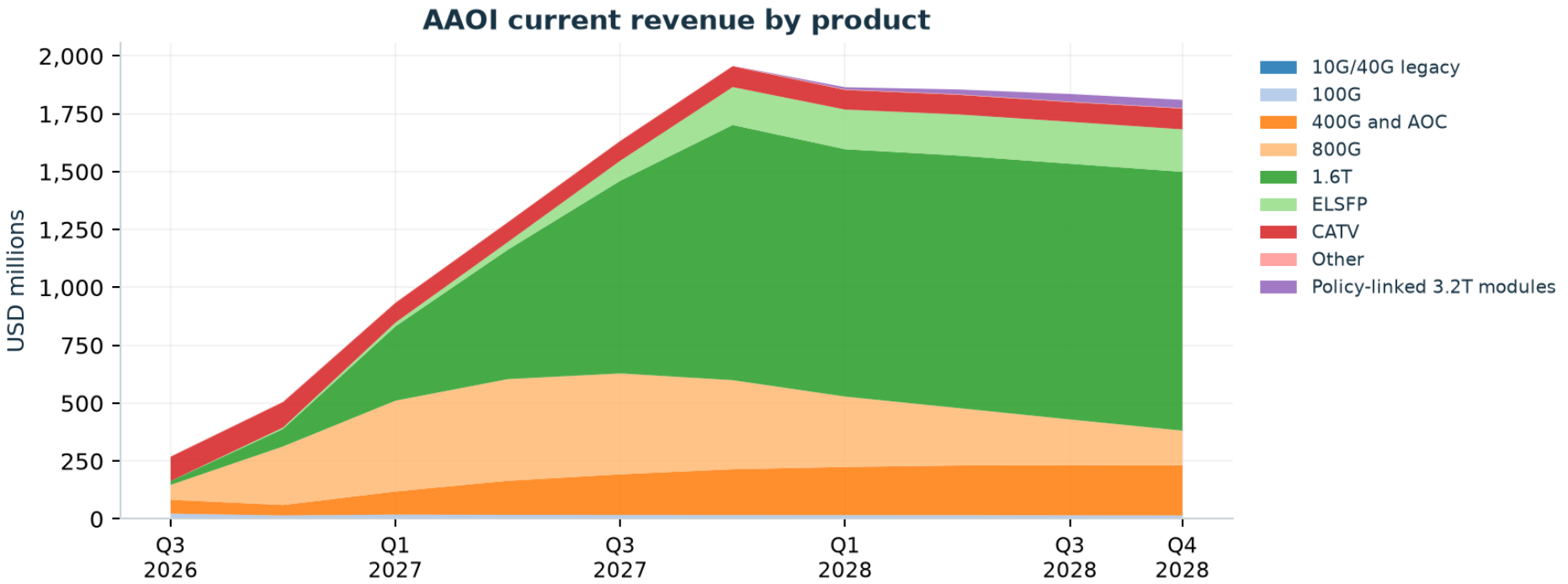


Figure 4. Current quarterly revenue by product, USD millions. All inherited product vectors remain; AAOI alone adds a separately labeled substitution vector.

Quarter	Revenue	Gross profit	COGS
Q3 2026	270.639	82.085	188.554
Q4 2026	506.744	172.175	334.569
Q1 2027	935.164	341.777	593.387
Q2 2027	1,283.068	494.048	789.020
Q3 2027	1,634.329	664.319	970.009
Q4 2027	1,957.352	830.706	1,126.646
Q1 2028	1,864.659	805.230	1,059.429
Q2 2028	1,855.014	809.476	1,045.538

Quarter	Revenue	Gross profit	COGS
Q3 2028	1,835.485	802.516	1,032.970
Q4 2028	1,809.987	787.780	1,022.207

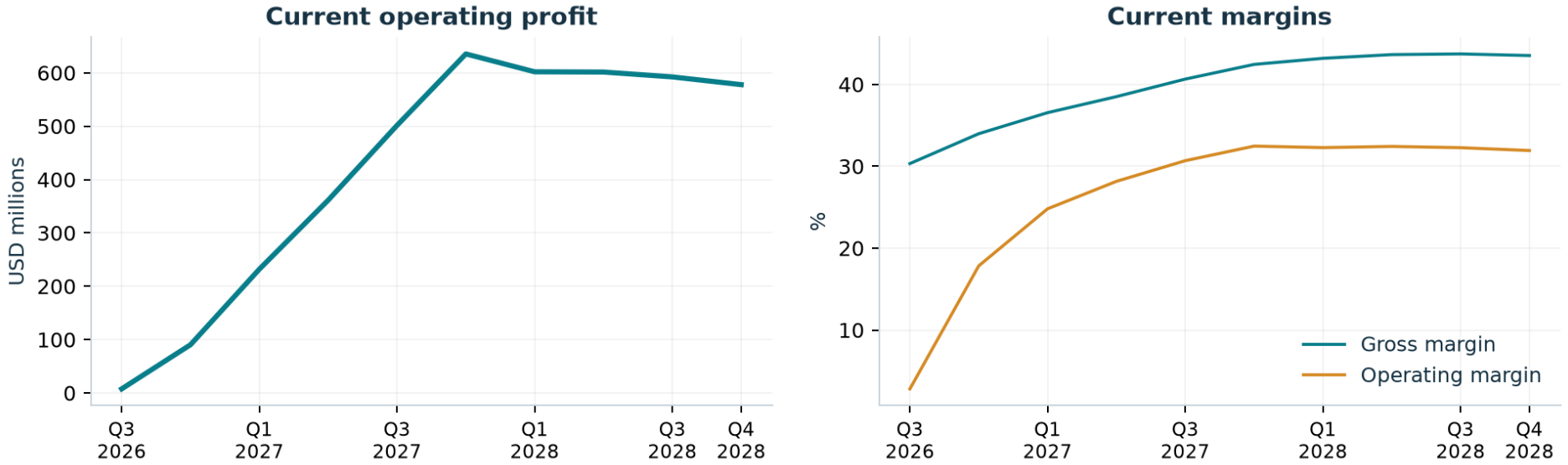


Figure 5. Current operating profit and margins. Operating profit is consolidated USD millions; the valuation uses parent-economic OP only where the inherited method already requires it.

Quarter	Revenue	Gross margin	Opex	Operating profit	OP margin	Valuation OP
Q3 2026	270.639	30.33%	74.480	7.605	2.81%	7.605
Q4 2026	506.744	33.98%	81.788	90.387	17.84%	90.387
Q1 2027	935.164	36.55%	109.806	231.970	24.81%	231.970
Q2 2027	1,283.068	38.51%	132.769	361.279	28.16%	361.279
Q3 2027	1,634.329	40.65%	162.873	501.447	30.68%	501.447
Q4 2027	1,957.352	42.44%	195.308	635.398	32.46%	635.398
Q1 2028	1,864.659	43.18%	203.339	601.891	32.28%	601.891
Q2 2028	1,855.014	43.64%	208.033	601.444	32.42%	601.444

Quarter	Revenue	Gross margin	Opex	Operating profit	OP margin	Valuation OP
Q3 2028	1,835.485	43.72%	210.192	592.324	32.27%	592.324
Q4 2028	1,809.987	43.52%	209.976	577.804	31.92%	577.804

Every revenue vector and product margin

Product GP = product revenue x the displayed modeled margin. Consolidated gross profit is the sum of the AAOI product gross-profit amounts shown below. Product allocations and future margins remain LOW-confidence model assumptions unless specifically identified as historical or guided.

10G/40G legacy

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	4.000	12.10%	0.484
Q4 2026	3.000	12.20%	0.366
Q1 2027	4.000	12.30%	0.492
Q2 2027	3.000	12.30%	0.369
Q3 2027	3.000	12.40%	0.372
Q4 2027	3.000	12.30%	0.369
Q1 2028	3.000	12.30%	0.369
Q2 2028	3.000	12.20%	0.366
Q3 2028	3.000	12.20%	0.366
Q4 2028	3.000	12.10%	0.363

100G

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	18.984	20.20%	3.835
Q4 2026	12.012	20.50%	2.462
Q1 2027	14.700	20.70%	3.043
Q2 2027	14.700	20.90%	3.072
Q3 2027	14.280	21.00%	2.999
Q4 2027	13.860	20.90%	2.897
Q1 2028	13.440	20.80%	2.796
Q2 2028	13.020	20.70%	2.695
Q3 2028	12.600	20.60%	2.596
Q4 2028	12.180	20.40%	2.485

400G and AOC

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	60.014	31.00%	18.604
Q4 2026	45.157	32.50%	14.676
Q1 2027	99.900	33.60%	33.566
Q2 2027	147.247	34.30%	50.506
Q3 2027	175.580	34.80%	61.102
Q4 2027	197.904	35.00%	69.266
Q1 2028	208.118	35.10%	73.050
Q2 2028	214.816	35.10%	75.400
Q3 2028	217.172	35.00%	76.010
Q4 2028	216.709	34.80%	75.415

800G

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	64.010	26.00%	16.642
Q4 2026	252.958	33.00%	83.476
Q1 2027	392.205	35.00%	137.272
Q2 2027	438.970	36.50%	160.224
Q3 2027	435.618	38.00%	165.535
Q4 2027	384.609	39.20%	150.767
Q1 2028	303.910	39.80%	120.956
Q2 2028	248.378	40.00%	99.351
Q3 2028	196.885	39.70%	78.163
Q4 2028	149.431	39.20%	58.577

1.6T

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	16.632	31.00%	5.156
Q4 2026	75.617	38.00%	28.734
Q1 2027	321.159	40.00%	128.464
Q2 2027	559.150	41.50%	232.047
Q3 2027	832.321	43.00%	357.898
Q4 2027	1,102.289	44.20%	487.212
Q1 2028	1,068.466	44.80%	478.673
Q2 2028	1,090.514	45.00%	490.731
Q3 2028	1,104.663	44.70%	493.784
Q4 2028	1,117.881	44.20%	494.104

ELSFP

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	0.000	50.00%	0.000
Q4 2026	6.000	50.00%	3.000
Q1 2027	16.200	50.50%	8.181
Q2 2027	33.000	51.20%	16.896
Q3 2027	86.530	52.26%	45.224
Q4 2027	163.690	53.33%	87.288
Q1 2028	171.167	54.38%	93.085
Q2 2028	177.700	55.24%	98.159
Q3 2028	181.080	55.83%	101.106
Q4 2028	183.494	56.03%	102.819

CATV

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	105.000	35.20%	36.960
Q4 2026	110.000	35.50%	39.050
Q1 2027	85.000	35.70%	30.345
Q2 2027	85.000	35.90%	30.515
Q3 2027	85.000	36.20%	30.770
Q4 2027	90.000	36.10%	32.490
Q1 2028	85.000	36.00%	30.600
Q2 2028	85.000	35.90%	30.515
Q3 2028	85.000	35.80%	30.430
Q4 2028	90.000	35.60%	32.040

Other

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	2.000	20.20%	0.404
Q4 2026	2.000	20.50%	0.410
Q1 2027	2.000	20.70%	0.414
Q2 2027	2.000	20.90%	0.418
Q3 2027	2.000	21.00%	0.420
Q4 2027	2.000	20.90%	0.418
Q1 2028	2.000	20.80%	0.416
Q2 2028	2.000	20.70%	0.414
Q3 2028	2.000	20.60%	0.412
Q4 2028	2.000	20.40%	0.408

Policy-linked 3.2T modules

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	0.000	0.00%	0.000
Q4 2026	0.000	0.00%	0.000
Q1 2027	0.000	0.00%	0.000
Q2 2027	0.000	0.00%	0.000
Q3 2027	0.000	0.00%	0.000
Q4 2027	0.000	0.00%	0.000
Q1 2028	9.558	55.30%	5.286
Q2 2028	20.586	57.54%	11.844
Q3 2028	33.085	59.39%	19.648
Q4 2028	35.291	61.12%	21.570

Full price target derivation

Valuation OP equals consolidated OP except where the saved model allocates economics to parent shareholders. The current model adds ownership x incremental OP to the saved parent-economic OP. Existing parent-only expenses and senior claims remain deducted as before. For every price target, sum that quarter and its next three quarters, multiply by the sanctioned multiple and currency/unit factor, and divide by the saved quarter-specific share denominator. No stock-price-dependent upside or ranking appears in this artifact.

price target quarter / window	F4Q valuation OP	Multiple	Shares M	FX / unit factor	price target
Q3 2026 to Q2 2027	691.241378	20x	92.800000	1.000000	148.974435
Q4 2026 to Q3 2027	1,185.082521	20x	96.000000	1.000000	246.892192
Q1 2027 to Q4 2027	1,730.094031	20x	99.000000	1.000000	349.513946
Q2 2027 to Q1 2028	2,100.014255	20x	101.000000	1.000000	415.844407
Q3 2027 to Q2 2028	2,340.179064	20x	103.000000	1.000000	454.403702
Q4 2027 to Q3 2028	2,431.056344	20x	105.000000	1.000000	463.058351
Q1 2028 to Q4 2028	2,373.462181	20x	106.000000	1.000000	447.823053

Q4 2027 worked example: (635.398057 + 601.890698 + 601.443666 + 592.323923) x 20 x 1 / 105.000000 = **463.058351 USD/share**.

Annual cross-check

Calendar-year sums below include only the quarters in the saved model. Partial years are labeled. These are not additional revenue, a horizon roll, or new management guidance.

Calendar year	Coverage	Revenue	Gross profit	Operating profit	OP margin
2026	2 quarters	777.383	254.260	97.992	12.61%
2027	4 quarters	5,809.912	2,330.850	1,730.094	29.78%
2028	4 quarters	7,365.145	3,205.002	2,373.462	32.23%

Probability and dilution sensitivity

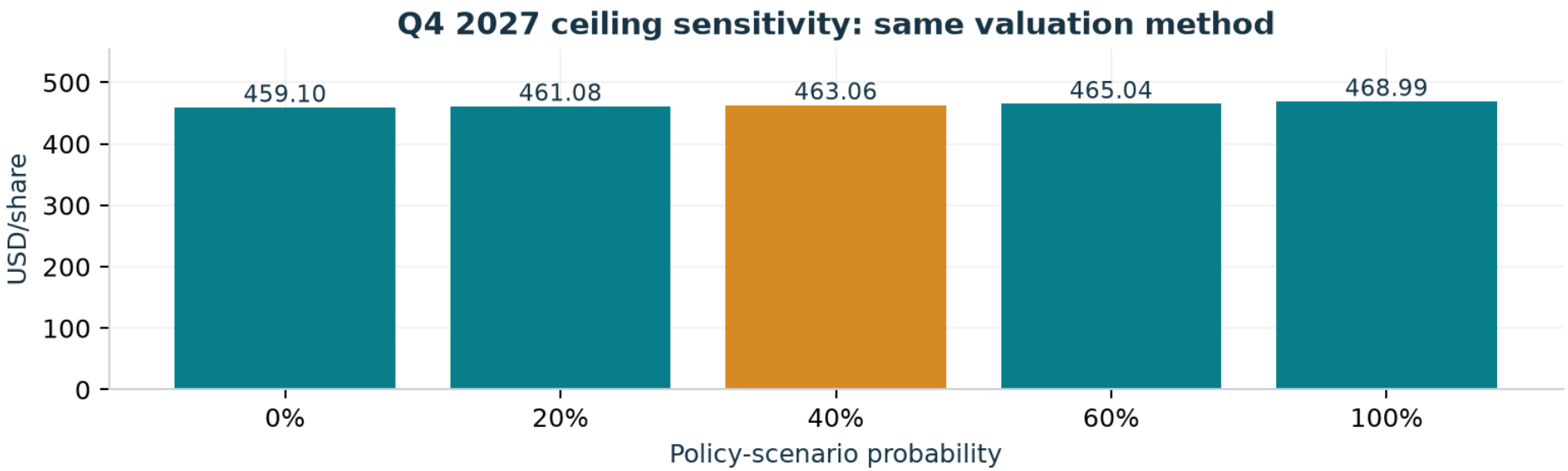


Figure 6. Q4 2027 price target under alternative policy probabilities. Gold is the selected 40% weight. All company valuation methods, operating reference inputs and shares remain unchanged.

Policy probability	Q4 2027 price target	Change from zero-policy sensitivity
0%	459.1022	0.000%
20%	461.0803	0.431%
40%	463.0584	0.862%
60%	465.0364	1.293%
100%	468.9926	2.154%

Additional diluted shares	Q4 2027 price target	Interpretation
0%	463.0584	Same operating case; denominator sensitivity only
5%	441.0080	Same operating case; denominator sensitivity only
10%	420.9621	Same operating case; denominator sensitivity only

The 40% weight is a judgment about the incremental policy economics and timing, not a statistical certainty. Because the nine companies share the same policy event, these increments should not be treated as nine independent sources of portfolio diversification. The next 12-16 months have the highest forecasting relevance; later continuation assumptions are progressively less reliable.

Validation and next evidence

This current artifact and the master model use the same selected 40% policy case. Existing company valuation methods, quarter conventions, ownership and dilution schedules are retained. Every existing portfolio Excel Base is protected; the analytical score above is assessed separately.

The incremental ledger exposes revenue, production costs, gross profit and operating expense. A zero-policy sensitivity removes only that ledger's contribution. It is a diagnostic control, not a competing active forecast. Reported facts and dated guidance retain their original dates; new pricing, exposure, costs and timing are labeled analyst assumptions.

The most useful next evidence is the proposed content definition, product-specific fabrication and assembly routes, contract repricing rights, delivered 3.2T qualification, accepted component premiums and unit-cost/yield data. Those observations can raise or lower the inputs and their probability; official confirmation is not a prerequisite for modeling this credible breadcrumb.

Current model: **AAOI_Q_Model** in **v3_2_master_portfolio.xlsx**. The full quarterly model includes all original product vectors and the approved incremental operating bridge. Approval snapshots and change history are retained externally rather than as competing current forecasts.